

Relational Model & SQL Fundamentals - Quick Revision Notes

QUICK REVISION & EXAM CHEAT SHEET

Subject: Database Management Systems | Chapter: Chapter 1: Relational Model, ER Diagrams & SQL Fundamentals | Academic Year: 2023-2024

Key DBMS Definitions

- * Primary Key: Unique, non-null identifier for table tuples.
- * Foreign Key: Attribute enforcing referential integrity by referencing primary key of another table.
- * Candidate Key: Minimal superkey capable of uniquely identifying tuples.
- * Superkey: Any set of attributes that uniquely identifies a row in a relation.

SQL Query Clause Order

- * Syntax Order: SELECT -> FROM -> WHERE -> GROUP BY -> HAVING -> ORDER BY.
- * Execution Order: FROM -> WHERE (row filter) -> GROUP BY -> HAVING (aggregate filter) -> SELECT -> ORDER BY.
- * WHERE filters individual rows before aggregation; HAVING filters groups after GROUP BY.

Join Types Quick Reference

- * INNER JOIN: Intersection of both tables matching join condition.
- * LEFT JOIN: All rows from left table + matched rows from right table (nulls if no match).
- * CROSS JOIN: Cartesian product of both tables (size = M * N rows).